Sahaj Bhadja

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Summary

Software Engineer with 2+ years building **Python** backend platforms, **async job orchestration** with explicit state machines, **event-driven** integrations, and CI/CD infrastructure automation on **Kubernetes**, Docker, AWS, and Azure.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Restructured fault-tolerant asynchronous job orchestration with FastAPI workers, explicit **job-state transitions**, failure recovery, and isolation, processing 1K+ AI jobs at 3x throughput without cascading failures under concurrent production load.
- Built **reconciliation-driven** integration pipelines across PostgreSQL, SQL Server, and cloud storage with validation checks and **CI/CD-backed releases**, sustaining 5K+ monthly jobs at 99%+ production reliability while cutting latency 35%.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Integrated OpenAI, Anthropic, and Qwen model providers behind one **switchable backend interface** with OCR fallback routing and timeout handling, serving 150+ instructors and 10K+ daily grading requests on AWS.
- Designed a **concurrent batch-processing engine** using ThreadPoolExecutor with fault-tolerant execution and per-call token cost tracking, processing 1K+ files per batch while reducing feedback turnaround 60% under peak load.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Automated a nightly **VM provisioning and test lifecycle** in Jenkins that allocated VMs, built and installed executables, and ran 20+ unit and integration suites across Windows, Linux, and macOS.
- Redesigned **microservice** API integrations and ETL workflows on AWS Glue and S3 with pagination, caching, and factory patterns, improving response times 30% and pipeline efficiency 68% while cutting technical debt 20%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a Node.js and Express **microservices** platform with 8 services for authentication, resource allocation, billing, and notifications, pushing real-time WebSocket updates while syncing 2M+ records across locations reliably.
- Automated Jenkins **CI/CD** that packaged services as Docker images, provisioned staging through ARM templates, and shipped zero-downtime **Kubernetes rolling updates**, using monitoring analytics to reduce resource wastage 42%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built Python Flask booking and payment APIs with **idempotent webhook handling** and transactional slot locking to prevent double bookings, keeping sub-200ms responses across 50K+ booking records and payment events.
- Engineered **event-driven** refund and dispute automation from payment webhooks plus hourly Spark micro-batches on Databricks, reducing manual refund handling 70% and pushing real-time status updates through Firebase to mobile clients.

Skills

Languages: Python, Java, C++, TypeScript, JavaScript, SQL, Bash, YAML
Platform: Kubernetes, Docker, State Machines, Async Orchestration, Reconciliation, Microservices, REST APIs, FastAPI, Flask
Reliability: Event-Driven Systems, Webhooks, WebSockets, Idempotency, Fault Isolation, Failure Recovery, Concurrency, Spark
Cloud & Data: AWS (EC2, S3, RDS, Glue), Azure, GCP, ARM Templates, Linux, PostgreSQL, SQL Server, MongoDB
Engineering Practices: CI/CD, Jenkins, GitHub Actions, PyTest, Integration Testing, Monitoring, Distributed Systems

Projects

Declarative Node Lifecycle Operator | *Python, Kubernetes CRDs, Temporal, PostgreSQL, Docker, PyTest*

- Designed **Kubernetes operator** with CRDs modeling host lifecycle from discovery through driver validation, health checks, drain, and decommission as versioned states on **durable Temporal workflows** with idempotent retries and drift detection.

GPU Capacity Rebalancer | *Python, Kubernetes, NATS, Prometheus, Grafana, Docker*

- Designed an **event-driven** capacity controller consuming NATS node-health events to cordon and drain failed GPU nodes, then applying **bin-packing and defragmentation** rebalancing to consolidate scattered workloads into contiguous capacity.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB, WebSockets, CI/CD*

- Deployed a fintech platform on **Kubernetes** with automated CI/CD, Plaid bank linking, token-based firewall access, and WebSocket real-time updates, supporting 50,000+ users and 1M+ transactions with scalable MongoDB access.